

Statistical Machine Learning Model for Stochastic Optimal Planning of Distribution Networks Considering a Dynamic Correlation and Dimension Reduction

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Abstract—Distributed generation and reactive power resource allocation will affect the economy and security of distribution networks. Deterministic scenario planning cannot solve the problem of network uncertainties, which are introduced by intermittent renewable generators and a variable demand for electricity. However, stochastic programming becomes a problem of great complexity when there is a large number of scenarios to be analyzed and when the computational burden has an adverse effect on the programming solution. In this paper, statistical machine learning theories are proposed to quickly solve the optimal planning for capacitors. Various technologies are used: Markov chains and copula functions are formulated to capture the variability and correlation of weather; consumption behavior probability is involved in the weather-sensitive load model; nearest neighbor theory and nonnegative matrix decomposition are combined to reduce the dimensions and scenario scale of stochastic variables; the stochastic response surface is used to calculate the probabilistic power flow; and probabilistic inequality theory is introduced to directly estimate the objective and constraint functions of the stochastic programming model. The effectiveness and efficiency of the proposed method are verified by comparing the method with the scenario reduction algorithm and the Monte Carlo method in a 33-bus distribution system.

Index Terms—Photovoltaic, capacitor, power grid planning, probabilistic power flow, statistical machine learning.

I. INTRODUCTION

POWER grid planning is an important part of power system planning. Its purpose is to determine the optimal grid structure for load growth during the planning period and a power planning scheme that can meet the demand for the economical and reliable transmission of electricity. In recent years, because of the interconnections between new energy sources, such as large-scale wind and solar energy, and stochastic factors, such as energy storage and weather-sensitive

loads, a large amount of uncertainty and stochastic behavior has emerged in power systems' sources. Thus, stochastic optimal planning has become an essential subject of research in power systems. Stochastic programming methods should be applied instead of deterministic models when the power flow in a power grid changes from deterministic power flow to probabilistic power flow (PPF). This change greatly interferes with the accuracy of the PPF, so it is necessary to improve the stochastic optimal planning method of a power grid with a high proportion of stochastic factors.

To more accurately study the influence of uncertainties on power systems, Borkowska first presented PPF theory in 1974 [1]. PPF uses statistical knowledge to describe the uncertainty in system state variables; PPF calculation methods can be divided into three categories: simulation methods, analytical methods and approximation methods. Monte Carlo simulation is a typical simulation method. Because its calculation accuracy depends on the sampling scale, it has the shortcomings of requiring heavy calculations and being time-consuming. However, its convergence and accuracy are not affected by the scale and complexity of the power system. The Monte Carlo method is usually used as the reference standard by which to measure other PPF calculation methods. The Nataf transform can solve the model problem of correlated nonnormal distributions, and Latin hypercube sampling can improve the efficiency of Monte Carlo sampling for PPF calculations [2]. Fu *et al.* used a copula function to simulate the correlation between photovoltaic (PV) power and heating, ventilating, and air-conditioning (HVAC) loads, and the influence of the thermoelectric correlation on the PPF was analyzed in a real 41-bus distribution system in China [3]. The analytical method mainly includes the convolution method, the point estimation method (PEM) and the semi-invariant method. The convolution method obtains the probability density function of the state variables via the convolution operation of the probability density function of the input stochastic variables [4], but the derivation of this method involves a great deal of mathematics. The PEM method does not require knowledge of the probability density function of the input stochastic variables and uses a Gram-Charlier expansion to approximate the expected value, variance and cumulative distribution of state variables [5]. The semi-invariant method obtains semi-invariants of the state variables from semi-invariants of input variables through algebraic

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operations [6] and requires linearization of the power flow equation. For the approximation method, an agent model can replace the actual PPF; this model has a high calculation accuracy and efficiency. In [7], the response surface is used to represent the thermoelectric hybrid PPF calculation with a good accuracy and efficiency. However, the response surface method cannot guarantee convergence by approximating the stochastic response of the structure with polynomials. The stochastic response surface method (SRS) overcomes this disadvantage and shows good accuracy and a high efficiency in PPF calculations [8]. In summary, the existing statistical-based PPF calculation model can address multidimensional stochastic variables, but the methods available for dealing with high-dimensional stochastic variables are not sufficiently mature. In addition, more research is needed on the ability to generalize PPF calculations based on agent models.

Stochastic planning is an important issue in distribution networks, as it is a matter of the utmost importance to ensure the safe and economic operation of a power grid with a high proportion of new energy sources. Research has been conducted in this field of inquiry. Fu *et al.* presented a distribution network planning model based on chance-constrained programming that realizes the coordinated planning of photovoltaic power generation and reactive power devices, and considered the balance of power quality and network loss when the grid is running [9]. Zhang *et al.* presented a linearized approximation algorithm to solve the interval reactive power optimization model, and the first-order terms of the Taylor expansions of the interval functions were utilized to approximate the interval reactive power optimization model [10]. An interval sequential quadratic programming algorithm that provided a new method for solving the interval dynamic reactive power optimization model was presented in [11]. It has a strong robustness and engineering application ability. Dynamic travel behaviors and the charge state of electric vehicles have a detrimental effect on the optimal solution for the capacity and configuration of electric vehicle charging stations. Based on the statistics of plug-in electric vehicles, Yang *et al.* used Markov chains and queuing theory [12] to present an optimal planning sizing model of fast-charging stations. Ehsan and Yang harnessed heuristic moment matching to characterize the uncertainty of representative scenarios [13]; their method can coordinately plan intermittent renewable generators. Optimal planning research has been conducted in the field of demand response: [14] emphasized that energy planning in heating/cooling systems management can minimize costs and improve reliability in smart microgrids. It is imperative that we provide power from a neighboring feeder or a distributed generation when a fault occurs in a feeder, and service restoration requirements should include optimal infrastructure planning of active distribution networks [15]. Optimal planning of PV power has been reinforced by extensive research, and both region-specific solar power and temperature should be formulated for the optimal planning of PV generation in low voltage microgrids [16]. Chronological time-period clustering can be used to determine the capacity expansion plans in electricity networks and can overcome the problem where time period aggregation approaches fail to properly capture the

mid-term dynamics of intermittent renewable generators [17]. In summary, the impact of stochastic characteristics introduced by intermittent renewable generators on power grid planning has been taken into account; however, it is still necessary to achieve a coordinated balance between speed and accuracy in stochastic optimization models with a large number of scenarios.

The novel contributions of this paper can be summarized as follows. 1) The copula function and Markov chain theory are used to model weather factors, considering not only cross-correlation but also state transition probabilities. The accurate modeling of weather factors ensures the accuracy of weather-sensitive load simulations and intermittent renewable generators. 2) The accuracy of the correlation among sources and loads is the basis for the accurate calculation of PPF. The total correlation is proposed as a measure for evaluating the correlation of more than two variables, and a Monte Carlo method is used to address the fact that the total correlation of multiple variables is difficult to calculate. 3) Large scenario datasets and a high-dimensional uncertainty make calculating the PPF complex and time-consuming. To avoid these drawbacks, a novel method based on the nearest neighbor approach and nonnegative matrix factorization (NMF) is proposed to produce SRS input variables with low dimensionality and a small size; the proposed PPF fast calculation method has a good efficiency and accuracy. 4) The Chebyshev inequality probability bound is then introduced to directly apply the PPF to the constraints of the stochastic optimal planning model, which can be efficiently solved via the proposed PPF fast calculation method.

The remainder of this paper is organized as follows. First, the uncertainty model of the source and load is described. Second, a fast calculation method is presented to solve the PPF calculation problems. Third, a stochastic planning model is presented to solve the optimal planning problems in distribution networks. Finally, four cases are simulated to verify the effectiveness of the proposed method.

II. GLOBAL VIEW OF THE PROPOSED MODEL

Statistical machine learning has a unique definition, which involves science about the probability and statistics model based on data, and it is then used to predict and analyze data. It is an interdisciplinary subject, involving probability theory, statistics, approximation theory, convex analysis, algorithm complex theory and other disciplines. Statistical machine learning is not only the core issue in the field of artificial intelligence; it has also been one of the most active research branches in the field of smart grid in recent years.

The motivation in this paper is to improve the efficiency of calculating PPF via statistical machine learning, which enables further accelerating the solution of stochastic optimal planning on distribution networks. This section focuses on a comprehensive view of the statistical machine learning methodology by highlighting the reasons and goals of addressing the problem with stochastic optimal planning of distribution networks. A method that integrates the statistical machine learning models is proposed in this paper, and it uses a set of submodel methods as follows.

TABLE I
RELATIONSHIP BETWEEN UNCERTAINTY MODEL AND
STATISTICAL MACHINE LEARNING

Submodel	Goals
Distribution function	Model weather uncertainty
Distribution law	Model residential HVAC electricity consumption behavior
Copula function	Model correlation between solar radiation and temperature
Markov chain	Model capricious weather scenarios considering dynamic correlation
Mutual information	Verify the HVAC and PV models

TABLE II
RELATIONSHIP BETWEEN PPF AND STATISTICAL MACHINE LEARNING

Submethod	Goals
Nearest neighbor approach	Determine the intrinsic dimensionality
NMF	Develop a mathematical formalism of dimensionality reduction
Reduction method	Reduce the dimension and scale of the high-dimensional input data
SRS	Compute PPF quickly based on small-scale DPF calculations

(a) An uncertainty model is proposed to simulate and verify the random and fluctuating data of HVAC consumption and PV generation while considering the dynamic correlation in Section III. To achieve the above goal, distribution functions of continuous random variables, distribution laws of discrete random variables, the copula function, Markov chain, and mutual information are introduced and used in combination. The reasons for applying statistical machine learning to the uncertainty model in Section III is described in Table I.

(b) The fast PPF calculation method is proposed to improve the efficiency of the PPF calculation via statistical machine learning in Section IV. To achieve the above goal, a reduction method for variable dimensions and a scenario scale is proposed based on the nearest neighbor approach and NMF. The number of deterministic power flow (DPF) calculations is equal to the number of scenarios, so a scenario scale reduction leads to a reduction in the DPF calculation burden. The reasons for applying statistical machine learning to the PPL method in Section IV is described in Table II.

(c) A stochastic optimal planning model is proposed to improve the solution efficiency of stochastic optimal planning in Section V. To achieve the above goal, a probabilistic inequality theory is introduced and is combined with PPF and the planning solution.

The connection relationships among the submethods are as follows: Section III provides the simulation sample data of the power source and load under different weather scenarios; Section IV calculates PPF using the sampling data from Section III; and Section V solves the stochastic planning solution using the PPF results.

III. UNCERTAINTY MODEL OF THE SOURCE AND LOAD

First, we give a concrete explanation on the dynamic correlation. Second, we use the Markov chain and copula function

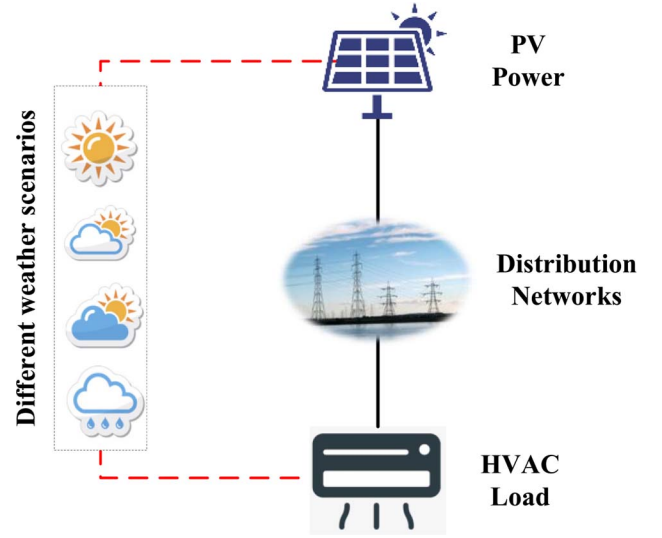


Fig. 1. Correlation between PV power and HVAC load.

to model the weather factors. Third, we establish a weather-sensitive load model that considers the probability of electricity consumption. Finally, we propose a solution method for the total correlation, which can be used to verify the accuracy of the correlated uncertainties, including weather sensitive loads and the PV power.

A. Dynamic Correlation

As shown in Fig. 1, both the HVAC load and PV power can be affected by fluctuating weather, so correlations exist between them. The correlation between the HVAC load and PV power was described in [3], and this paper considers different degrees of correlation under different weather scenarios. PV power is sensitive to solar radiation, and HVAC load is sensitive to temperature. The correlation degree between solar radiation and temperature is different under different weather scenarios, so the correlation degree between the HVAC load and PV power is also different under different weather scenarios. One can suppose that there is a type of correlation in a weather scenario, so dynamic correlation is needed to describe weather scenario changes. To describe the changing law of weather scenarios, the dynamic correlation between solar radiation and temperature will be modeled and further elaborated on in the next section.

B. Weather Model

Different probability distribution functions (PDFs) have evolved to adapt to changes in solar radiation along with changes in geographical area and the temporal resolution. In this article, we presume that the PDF of solar radiation is the same as the beta distribution in [18]:

$$PDF(G) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \left(\frac{G}{G_{\max}}\right)^{\alpha-1} \left(1 - \frac{G}{G_{\max}}\right)^{\beta-1}, \quad (1)$$

where Γ is the gamma function, α and β are the shape parameters, and G and $G_{\max}$ are the current and maximum solar radiation values, respectively.

Procedure 1 Process for Stochastic Weather Simulation

Step 1. Statistical calculation.

- Calculate the CDFs of the outdoor temperature and solar radiation using (1)-(2).

- Calculate the CDFs and matrix ρ of the copula using (3).

Step 2. Dynamic correlation modeling.

- Divide the copula CDFs into multiple intervals so that each interval contains the same number of original samples. Each interval is defined as a state.

- Create the Markov chain empirical probability of going to state(j) from state (i) via statistical calculations.

- Create empirical discrete distributions (of interval size dt) for each state.

Step 3. Weather simulation generation.

- Generate a sample state path using a Markov chain.

- Simulate a discrete copula CDF when in a particular state.

- Calculate the CDF of the outdoor temperature and solar radiation by calculating the inverse function of the discrete copula CDF.

- Generate the sample points for the outdoor temperature and solar radiation by calculating the inverse function of the weather marginal CDFs.

The PDF of the multidimensional outdoor temperature can be assumed to be a normal distribution [3]:

$$PDF(T) = \frac{1}{\sigma_{sh}\sqrt{2\pi}} \exp\left[-\frac{(T - \mu_{sh})^2}{2\sigma_{sh}^2}\right], \quad (2)$$

where T is the outdoor temperature and μ_{sh} and σ_{sh} are the mean and standard deviations, respectively.

A copula function exists with independent variables, which are the marginal cumulative probability distributions of the radiation and temperature:

$$Y = copulacdf(CDF(G), CDF(T), \rho), \quad (3)$$

where CDF ($\bullet$) is a cumulative distribution function, *copulacdf* ($\bullet$) is a CDF for a copula, and ρ is a matrix of linear correlation parameters.

The copula CDF is controlled to model the dynamic correlation among capricious factors in weather samples. The Markov chain model for regime-switching time series is built by using a stochastic generation to simulate the copula CDF by setting up several copula CDF regimes. Based on the formulas provided above, the method of stochastic simulation in accordance with realistic samples is given in Procedure 1.

Remark 1: Traditionally, the mutual correlation is fully modeled, but the state transition probability (i.e., the scenario dynamic change law) is neglected. In this paper, the Markov chain of a copula is modeled, and the dynamic correlation characteristics of the variables are determined to improve the accuracy of the weather-sensitive load calculations.

C. PV and HVAC Model

PV power and the weather-sensitive load are the two main uncertainties considered in this paper, and we mainly introduce the weather-sensitive load. The PV model in [19] is introduced to generate PV power samples, so a PV model is not introduced here. Additionally, the PV active power output is a function of temperature and solar radiation. The HVAC model in [20] is introduced to generate HVAC load samples, and our contribution is to propose an HVAC thermostat setting model by simulating the residential behavior probability.

Procedure 2 Process for Stochastic Simulation of m Thermostat Setpoints

Step 1. Obtain the distribution law of the thermostat setpoints.

Step 2. Divide the range $[0, 1]$ into k bins, where the length of each bin is proportional to the probability mass.

Step 3. Randomly generate m values from a uniform distribution in $[0, 1]$.

Step 4. The bins containing these values are selected as the probability array, whose corresponding discrete random variables are n simulated thermostat setpoints.

An equivalent thermal parameter of a residential HVAC unit can be expressed as follows:

$$T_{room}^{t+1} = T_a^{t+1} + QR - (T_a^{t+1} + QR - T_{room}^t)e^{-\Delta t/RC}, \quad (4)$$

$$T_{room}^{t+1} = T_a^{t+1} - (T_a^{t+1} - T_{room}^t)e^{-\Delta t/RC}, \quad (5)$$

$$P_{HVAC} = s_{HVAC} \times Q \div COP, \quad (6)$$

where T_{room} is the indoor temperature; Q is the equivalent heat rate; C is the equivalent heat capacity of a cooling area; R is the equivalent thermal resistance; t is time (min); Δt is the time step, which is set to one minute; P_{HVAC} is the power consumption of an HVAC; COP is the ratio of the thermal energy to power consumption; and s_{HVAC} stands for the switching state. Eq. (4) models the thermal state when the HVAC is on, Eq. (5) models the thermal state when the HVAC is off, and Eq. (6) represents the HVAC power consumption function.

HVAC units incorporate a deadband to prevent constant switching. This deadband largely drives the dynamics of the HVAC loads:

$$\begin{cases} s_{HVAC}(t) = 1, T_{room} \geq T_{upper} \\ s_{HVAC}(t) = s_{HVAC}(t-1), T_{upper} > T_{room} > T_{lower} \\ s_{HVAC}(t) = 0, T_{room} \leq T_{lower}, \end{cases} \quad (7)$$

$$T_{upper} = T_{set} + T_{band}/2, \quad (8)$$

$$T_{lower} = T_{set} - T_{band}/2, \quad (9)$$

where T_{set} is the residential thermostat temperature setpoint, T_{band} is the HVAC deadband, T_{upper} is the upper limit, and T_{lower} is the lower limit. In this paper, the distribution law of discrete random variables is used to establish the probabilistic model of residential HVAC electricity consumption behavior. The discrete random variable is an integer representing a thermostat temperature setpoint, which ranges from 16°C to 30°C. The distribution law of the thermostat temperature setpoints is calculated by statistical samples. The process for simulating the thermostat setpoint is in Procedure 2.

Remark 2: Note that the accuracy of the PPF calculation is related to the correlation. Markov theory for both weather and residential HVAC consumption behavior is proposed to ensure that the correlation is in line with reality.

D. Total Correlation

The correlations among the intermittent renewable generators and the electricity demands must be checked to ensure accurate PPF results; the total correlation is introduced to measure information that more than two variables have in common. Let $[p_1, p_2, \dots, p_d]$ denote the HVAC loads and PV active power outputs. The multidimensional joint entropy can

be expressed as

$$H = - \int_0^\infty \cdots \int_0^\infty \text{PDF}(p_1, p_2, \dots, p_d) \times \log[\text{PDF}(p_1, p_2, \dots, p_d)] dp_1, dp_2, \dots, dp_d. \quad (10)$$

The joint copula entropy can be calculated using the d univariate marginal entropies and the copula entropy as follows:

$$H(p_1, p_2, \dots, p_d) = \sum_{i=1}^d H(p_i) + H_C(u_1, u_2, \dots, u_d), \quad (11)$$

$$H_C(u_1, u_2, \dots, u_d) = - \int_0^1 \cdots \int_0^1 \text{PDF}(u_1, u_2, \dots, u_d) \times \log[\text{PDF}(u_1, u_2, \dots, u_d)] \times du_1, du_2, \dots, du_d, \quad (12)$$

where $H(\bullet)$ stands for the marginal entropy function, $[u_1, u_2, \dots, u_d]$ are uniformly distributed random variables, and $H_C(\bullet)$ is the copula entropy function.

For a multivariate vector with support in $[0, 1]$, Eq. (12) can be expressed as

$$H_C = - \int_{[0,1]^d} \text{PDF}(u_1, u_2, \dots, u_d) \times \ln[\text{PDF}(u_1, u_2, \dots, u_d)] du_1, du_2, \dots, du_d = -E[\ln[\text{PDF}(u_1, u_2, \dots, u_d)]], \quad (13)$$

where $E(\bullet)$ stands for the mathematical expectation function, which is calculated using the Monte Carlo method rather than by mathematical reasoning; this explains why there is a convergence problem when calculating the total correlation in the following simulations.

According to the definition in [21], the total correlation can be expressed as

$$I = \sum_{i=1}^d H(p_i) - H(p_1, p_2, \dots, p_d). \quad (14)$$

By comparing Eqs. (11) and (14), we obtain the following:

$$I = -H_C(u_1, u_2, \dots, u_d). \quad (15)$$

Eq. (15) is used to calculate the total correlation among the renewable generators and demands, and the Monte Carlo method for (13) is the key technique for solving (14).

Remark 3: Mutual information can only calculate the correlation of two variables. To evaluate the correlation of multiple stochastic variables, a total correlation is introduced. Copula entropy can estimate the total correlation directly via Monte Carlo sampling and avoid accumulation of a systematic bias.

IV. FAST PPF CALCULATION METHOD

First, we propose a reduction method for the dimension and scale of high-dimensional input data (i.e., HVAC loads and PV power) based on the nearest neighbor approach and nonnegative matrix factorization. Second, a PPF calculation method based on SRSF is introduced.

Procedure 3 Procedure of Intrinsic Dimensionality

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- Step 1. Set a value for K .
 Step 2. Determine all of the distances r_{k,p_i} .
 Step 3. Remove outliers by keeping the points that satisfy (20)
 Step 4. Compute $\log(\bar{r}_k)$
 Step 5. Initialize the estimator n' by fitting a line to $\log(\bar{r}_k) = (1/n') \log(k)$
 Step 6. Compute $\log(G_{k,n'})$ using (19)
 Step 7. Update n' by fitting a line to $\log(G_{k,n'}) + \log(\bar{r}_k) = (1/n') \log(k)$
 Step 8. Repeat steps 6 and 7 until the estimates converge.
-

A. Reduction of the Dimension and Scale

The nearest neighbor algorithm used here is not the original one; its function here is instead to determine the intrinsic dimensionality (sometimes called the effective dimensionality) of the input data. We derive a new series of dimension reduction formulas based on the intrinsic dimensionality and NMF theory. We estimate the intrinsic dimensionality using the algorithm in [22] based on the nearest-neighbor distances of a data set. We can suppose that the average k^{th} nearest neighbor distance is

$$\bar{r}_k = \frac{1}{d} \sum_{i=1}^d r_{k,p_i}, \quad (16)$$

where $r_{k,p}$ is the distance from p to the k^{th} nearest neighbor of p .

From this, it follows that the expected function is given by

$$E(\bar{r}_k) = \frac{1}{G_{k,n}} k^{1/n} C_d, \quad (17)$$

$$G_{k,n} = \frac{k^{1/n} \Gamma(k)}{\Gamma(k + 1/n)}, \quad (18)$$

$$C_d = \frac{1}{d} \sum_{i=1}^d [d \times p_i \times V_n]^{-1/n}, \quad (19)$$

$$V_n = \frac{(\pi)^{n/2}}{\Gamma(\frac{n}{2} + 1)}, \quad (20)$$

where n is the intrinsic dimension, C_d is sample-dependent, which is independent of k , V_n is the volume of the unit n -dimensional hypersphere, and p_i is the i^{th} density. The estimator n' can be estimated using:

$$\log(G_{k,n'}) + \log(\bar{r}_k) = (1/n') \log(k) + \log(C_d). \quad (21)$$

To ensure that the algorithm works well, outliers should be deleted by providing a reasonable range, as follows:

$$r_{k,p_i} \leq \frac{1}{d} \sum_{i=1}^d r_{K,p_i} + \sqrt{\frac{1}{d-1} \sum_{i=1}^d \left(r_{K,p_i} - \frac{1}{d} \sum_{i=1}^d r_{K,p_i} \right)^2}, \quad (22)$$

where K is the maximum number of nearest neighbors. The procedure of the intrinsic dimensionality is given in Procedure 3.

Next, we use the NMF in [23] in our development of the mathematical formalism of dimensionality reduction. NMF views matrix factorization as a constrained optimization problem that seeks to factor the original matrix V into the product of two nonnegative matrices W and A :

$$W \times A = V + \varepsilon, \quad (23)$$

$$p_i = F_i^{-1}(\Phi(V_i)), \quad (24)$$

where the physical meaning of V is the isoprobabilistic transformation of $[p_1, p_2, \dots, p_d]$; the size of V is $d \times l$; l denotes the sample scale, which is computed using SRSM; $\Phi(\bullet)$ is the standard normal cumulative distribution function; $F^{-1}(\bullet)$ is the estimated inverse cumulative distribution function based on kernel density estimates; the size of A is $n \times l$, where $n < d$; the size of W is $d \times n$; and ϵ is noise.

This paper resolves NMF by means of alternative non-negative least squares using projected gradients, and (23) is reformulated as:

$$\min_A \bar{f}(A) \equiv \frac{1}{2} \|V - WA\|_F^2 \text{ subject to } A_{bj} \geq 0, \forall b, j. \quad (25)$$

The columns of A are concatenated to form a vector $\text{vec}(A)$,

$$\bar{f}(A) = \frac{1}{2} \text{vec}(A)^T \begin{bmatrix} W^T W & & \\ & \dots & \\ & & W^T W \end{bmatrix} \text{vec}(A) + \text{term}(A), \quad (26)$$

where $\text{term}(\bullet)$ denotes the linear term, the Hessian matrix of (26) is block diagonal, and $W^T W$ is a positive semidefinite matrix. This matrix tends to be well-conditioned, so the gradient-based method can converge sufficiently quickly.

The low-dimensional data are obtained as follows:

$$X^T = A \times (V^T - \mu_V), \quad (27)$$

where μ_V stands for the mean of V and X is the matrix of low-dimensional data, whose size is $l \times n$. We eliminate the correlation of X using the orthogonal transform in [24]:

$$LL^T = \text{Cov}(X), \quad (28)$$

$$Z^T = L^{-1} X^T, \quad (29)$$

where Z is uncorrected data and L is a lower triangular matrix obtained via Cholesky factorization.

The matrix Z should be standardized to input the SRSM:

$$\xi = [Z - E(Z)] / \sqrt{D(Z)}, \quad (30)$$

where $E(\bullet)$ is the mean function, $D(\bullet)$ is the standard deviation function, and $\xi = \{\xi_i\}_{i=1}^n$ is the experimental data of the SRSM.

Usually, the SRSM needs to ensure that the random input variables are independent and the standard normal distributions are uncorrelated. How to meet the above requirements is described as follows: First, we transform the nonnormal distributions of the PV and HVAC data into standard normal distribution V using (24). Although solar radiation obeys a beta distribution, the HVAC load may not follow the same distribution type. That is, why we adopted kernel density estimates in (24). Second, we transform V into low-dimensional data X using (27). It is well known that the linear combination of d -dimensional standard normal components obeys a normal distribution, and the mathematical expectation of normal distribution is regarded as a position parameter. It can be concluded that X must obey a normal distribution, but it is no longer a standard normal distribution such as V . Third, we eliminate the correlation of X using (29) to obtain

the uncorrelated normal distribution Z . Finally, Z is standardized using (30) to obtain the SRSM's random input variables, which are independent and obey uncorrelated standard normal distributions.

Remark 4: Scenario reduction can prevent a decrease in the PPF calculation accuracy caused by large data fluctuation, and feature dimensionality reduction can effectively improve the efficiency of the PPF machine learning. By definition, the sample size reduction is a scenario reduction, and the dimensionality reduction is a mapping from a high-dimensional feature space to a low-latitude feature space.

B. PPF Calculation Method Based on SRSM

SRSM is an explicit polynomial chaos expansion based on Hermite polynomials [25]:

$$y = a_0 + \sum_{i_1=1}^n a_{i_1} \Gamma_1(\xi_{i_1}) + \sum_{i_1=1}^n \sum_{i_2=1}^{i_1} a_{i_1 i_2} \Gamma_2(\xi_{i_1}, \xi_{i_2}) + \sum_{i_1=1}^n \sum_{i_2=1}^{i_1} \sum_{i_3=1}^{i_2} a_{i_1 i_2 i_3} \Gamma_3(\xi_{i_1}, \xi_{i_2}, \xi_{i_3}) + \dots \quad (31)$$

where ξ_{i1} , ξ_{i1} and ξ_{i1} are standard normal distribution variables that are obtained via data processing of PV and HVAC samples; y is the response vector, which represents a state variable; $\Gamma_p(\bullet)$ denotes the multidimensional Hermite polynomials of order p ; and a_i is an unknown deterministic coefficient that needs to be estimated.

The approximation accuracy of (31) depends on p . The number of unknown deterministic coefficients is

$$n_a = \frac{(n+p)!}{n!p!}, \quad (32)$$

The sample size l is

$$l = 2 \times n_a. \quad (33)$$

The unknown coefficients in the polynomial chaos expansion can be determined by minimizing an appropriate norm of the residual [26], and we use the least squares method to structure a surface. Note that the sample size of the input variables is twice as large as the number of undetermined coefficients:

$$N_{\text{SRSM}} = n_a, \quad (34)$$

where N_{SRSM} is the sample size of the input variables, whose corresponding response vector is obtained using the power flow calculation as experimental data. When the input size is far less than the actual number of scenarios, it is obvious that we can improve the calculation speed of the PPF.

A second-order SRSM is introduced to estimate the PPF, and digital characteristics of the response vector can be calculated using

$$E(y) = a_0, \quad (35)$$

$$V(y) = \sum_{i=1}^K a_i^2 + 2 \sum_{i=1}^K a_{ii}^2 + \sum_{i=1}^{K-1} \sum_{j>i}^K a_{ij}^2, \quad (36)$$

where $E(\bullet)$ is the mathematical expectation function for the SRSM and $V(\bullet)$ is the variance function for the SRSM. By calculating the coefficients, we can not only obtain the probability

digital characteristics but also the responses corresponding to a group of inputs using (31).

We use the SRS to reduce the DPF calculations and scenario size, the purpose of which is to reduce the PPF time cost. The number of scenarios is also the sample size of the input data, and scale reduction is an important reason for the improvement in the PPF calculation's efficiency using the SRS. According to (31), the higher the dimension is, the more items there are in the formula. More items lead to a more complicated calculation process. The application condition of the SRS can be summarized as follows: (a) the time cost of the dimensionality reduction + the time cost of the SRS < the DPF time reduction accompanied by the sample size reduction. When the total time cost of the SRS with a certain low dimensionality and dimensionality reduction is equal to that of the reduced DPF calculations, that low dimensionality is the limitation for the SRS. (b) According to (33), the sample size needs to be twice the number of SRS coefficients to realize the SRS calculation. If the number of SRS coefficients is half of the sample size of the primal data, the sample size will not be further reduced. In this case, the number of scenarios is not reduced, and the workload of the DPF calculation is not reduced. Therefore, the premise of using the SRS to calculate the PPF is that the number of SRS coefficients is less than half of the original sample size. We can obtain the limitation for a low dimensionality according to the limitation for the number of SRS coefficients using (32). Considering the time cost of the dimensionality reduction and the SRS, the number of SRS coefficients should be much smaller than half of the original sample size.

Remark 5: Essentially, the SRS is an analytical method that combines statistics with mathematics. Compared with the academic idea of a scenario reduction, the SRS considers the probabilistic relationship between the input (uncertainty in the power supply and load) and the output (state variable), while the scenario reduction equates the probabilistic characteristics of the output with the probabilities of the scenarios.

V. STOCHASTIC OPTIMAL PLANNING

A. Model

Stochastic optimal planning has been an essential subject of research in distribution networks, and a stochastic programming model should be applied instead of deterministic models when intermittent renewable generators are developed. The economy objective function of the proposed programming model is

$$\min f_{\text{obj}}(S, \text{num}) = \text{mean}(p_{\text{loss}}), \quad (37)$$

where p_{loss} is the total active power loss in the distribution networks, S is the capacitor capacities, and the variable num denotes the capacitor bus number.

The constraints of the proposed programming model are

$$\begin{cases} S_{\min} \leq S \leq S_{\max} \\ 1 < \text{num} < \text{num}_{\text{sys}} \\ s_j < s_{\text{limit},j} \\ \Pr(v_{\min} < v_i < v_{\max}) \geq \alpha, \end{cases} \quad (38)$$

where $S_{\min}$ is the lower limit of the capacity; $S_{\max}$ is the upper limit of the capacity; num_{sys} is the number of buses; s_j is the transmission power of the j^{th} branch; $s_{\text{limit},j}$ is the limit of the j^{th} branch; v_i is the i^{th} bus voltage amplitude; $v_{\min}$ is the voltage lower limit; $v_{\max}$ is the voltage upper limit; and α is the minimum of the qualified voltage confidence level for capacitor buses.

Now we use the Chebyshev inequality in our development of (38) to further combine the PPF and stochastic optimal planning models:

$$\Pr(|v_i - E(v_i)| < \varepsilon) \geq 1 - \frac{V(v_i)}{\varepsilon^2}. \quad (39)$$

The inequality in (39) can be extended to the interval $[v_{\min}, v_{\max}]$:

$$\Pr(v_{\min} < v_i < v_{\max}) \geq 1 - \frac{V(v_i) + [E(v_i) - \frac{v_{\min} + v_{\max}}{2}]^2}{(\frac{v_{\max} - v_{\min}}{2})^2}. \quad (40)$$

Remark 6: Obviously, both (37) and (40) can be obtained using (35) and (36). The contribution of this paper is to use the fast PPF calculation method to improve the efficiency in planning solutions with the help of inequality theory. The planning model can be adjusted and improved according to actual engineering needs.

B. Conservative Discussion

It should be noted that the Chebyshev inequality is used for security constraints rather than the economy objective in this paper. If the objective function is too conservative it will reduce the economy performance, but this paper uses the mathematical expectation to describe the objective; for the safety constraints, if it meets the conservative condition it must also meet the safety constraints under the worst conditions. The conservative security constraints may reduce the feasible solutions, but this does not directly lead to a decline in the objective performance. Only when the eliminated feasible solution region includes the optimal solution does the objective performance decline.

The Chebyshev inequality is used in the stochastic optimal planning model, but in principle the constraint derived by the Chebyshev inequality is conservative. Monte Carlo can compare the Chebyshev inequality to address this issue. The Chebyshev inequality uses very little information, and only a mathematical expectation and variance are involved. The Chebyshev inequality tells us that the probability for a random variable deviating from its mean value is limited by its variance. When ε is smaller, the calculation results are more accurate. The Chebyshev inequality cannot be wrong; at most it is conservative. Monte Carlo has a better accuracy than the Chebyshev inequality, but the time cost of Monte Carlo is greater than that of the Chebyshev inequality.

Note that the SRS can improve the PPF efficiency by reducing the conventional DPF calculations accompanied by reducing the scenario size. There are a few scenarios needed for the SRS, so the sample size for the DPF calculations is small. There are not a large number of responses in the SRS

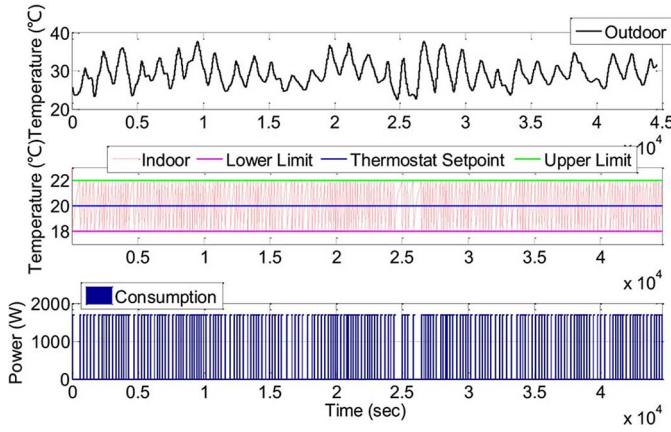


Fig. 2. Thermal behavior of an HVAC unit.

calculation results, but the mathematical expectation and variance are present. The application of Chebyshev inequality only requires the existence of a mathematical expectation and variance. However, Monte Carlo requires many computing resources. In essence, the Chebyshev inequality can estimate a reliable constraint level with only a small number of DPF calculations, and this advantage is why we chose the Chebyshev inequality.

VI. CASE STUDY AND DISCUSSIONS

We use a step-by-step validation method to demonstrate our example. First, we introduce the simulation results of a single HVAC model. Second, we analyze the uncertainty and correlation of HVAC loads at different buses in the distribution networks. Then, we give the calculation results for the PPF, and finally we give the results of reactive power planning. These cases are simulated using MATLAB. The machine performance will affect the simulation time. The simulation in this paper was performed at a Dell workstation, whose processor is an Intel Core i7-7820HQ CPU@2.90 GHz, with 31.9 GB of available memory.

A. HVAC Simulation

The simulation parameters of the thermal dynamic models of an HVAC unit are set according to those in Eq. (3) in [20], where the difference is that the number of cooling areas is set to fourteen, the customer thermostat setting is set at 20°C, and the thermostat deadband is set to 4°C. The ambient temperature in July is exported from a monitoring point in Beijing, China in order to simulate the HVAC unit's operation when the outdoor temperature changes. As shown in Fig. 2, the outdoor temperature is shown on top, the indoor temperature response with a certain thermostat deadband is shown in the middle, and the energy consumption is shown on the bottom.

By comparing Fig. 2 in this paper with Fig. 1 in [20], it can be concluded that the behavior pattern is reasonably established and that the simulation is in agreement with the actual engineering requirements. The simulations use a time scale at the minute level; the average power consumption of the HVAC is given in Fig. 3.

Discussion 1: Note that HVACs incorporate deadbands to prevent constant switching, and these deadbands largely drive

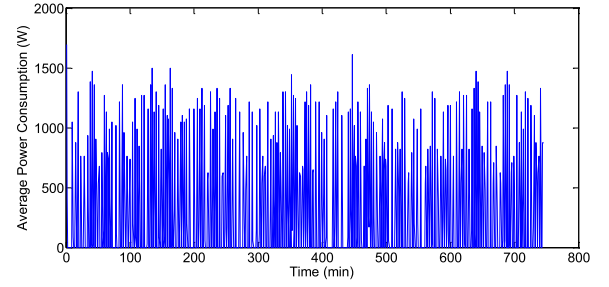


Fig. 3. Average power consumption of an HVAC unit.

TABLE III
PROBABILITY DISTRIBUTIONS FOR CUSTOMER THERMOSTAT SETPOINTS

Setpoint	Probability	Setpoint	Probability	Setpoint	Probability
16°C	0.01	21°C	0.025	26°C	0.2
17°C	0.01	22°C	0.025	27°C	0.025
18°C	0.01	23°C	0.2	28°C	0.025
19°C	0.01	24°C	0.2	29°C	0.025
20°C	0.01	25°C	0.2	30°C	0.025

the dynamics of the HVAC loads. It can be concluded that any HVAC current operating scenario will obviously be affected by the previous operating scenario. When establishing a weather-sensitive load model, the change rule of weather scenarios should be taken into consideration to ensure a good HVAC approximation simulation; therefore, we use a Markov chain to build a weather model.

B. Simulation of Correlated Loads

In the last section, we validated and analyzed the relationships between different operational scenarios of weather-sensitive loads. In this section, we validate the importance of dynamic correlation modeling by analyzing the uncertainty based on total correlation.

Because case33bw from MATPOWER, which is a free, open-source MATLAB-language package of M-files [27], is selected as the simulation object of this article, it is assumed that each HVAC load is connected to a PQ node and that there are ten HVAC units at each PQ bus. To model the uncertainty in the user behavior, it is assumed that customer thermostat settings obey the probability distributions listed in Table III.

In this case, three temperature zones are involved, and it is assumed that the PQ nodes are evenly distributed among the three regions according to their numbers. The three zone temperatures obey normal distributions with means [32.25°C, 30.44°C, 30.53°C] and standard deviations [3.74°C, 4.25°C, 2.56°C]. Solar radiation obeys a beta distribution with parameter estimates [0.043, 0.596]. The proposed method is compared with the traditional correlation model in [3] and with the raw data (i.e., actual results). The sample size is 500 for this data set. The Gaussian copula function and the total correlation are selected as evaluation indexes to analyze the dynamic correlation of the simulated weather and HVAC samples. In terms of physical meaning, the sample size and the number of scenarios are the same in this paper.

Both the proposed method and the traditional correlation mode in [3] use the Gaussian copula function to model the

TABLE IV
MATRIX OF LINEAR CORRELATION PARAMETERS
FOR A GAUSSIAN COPULA

Method	Weather	HVAC consumption
Proposed	[1.00, 0.36, 0.39, -0.16; 0.36, 1.00, 0.26, -0.15; 0.39, 0.26, 1.00, -0.09; -0.16, -0.15, -0.09, 1.00]	[1.00, 0.16, 0.14; 0.16, 1.00, 0.05; 0.14, 0.05, 1.00]
Real data	[1.00, 0.29, 0.3, 0.09; 0.29, 1.00, 0.12, 0.00; 0.35, 0.12, 1.00, 0.06; -0.09, 0.00, -0.06, 1.00]	[1.00, 0.11, 0.18; 0.11, 1.00, 0.08; 0.18, 0.08, 1.00]
Model in [3]	[1.00, 0.29, 0.32, 0.21; 0.29, 1.00, 0.06, 0.02; 0.32, 0.06, 1.00, 0.07; -0.21, 0.02, 0.07, 1.00]	[1.00, 0.11, 0.12 0.11, 1.00, 0.03 0.12, 0.03, 1.00]

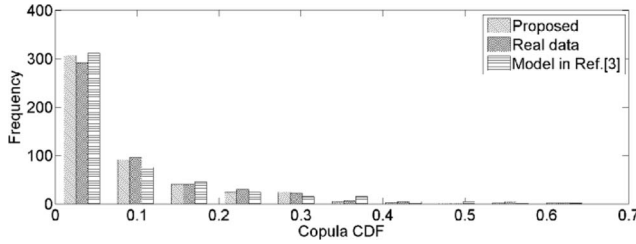


Fig. 4. Frequency of the Gaussian copula function for weather samples.

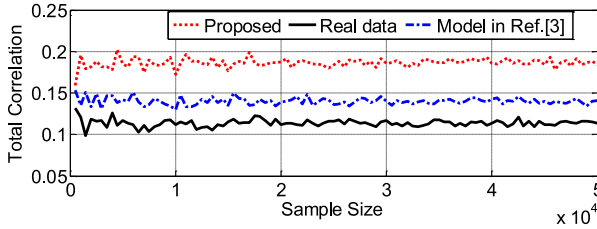


Fig. 5. Total correlation values of weather samples.

correlations among different uncertainties, such as the weather samples (temperature and solar radiation) and HVAC loads at different buses. The matrix of the linear correlation parameters for a Gaussian copula is estimated using the maximum likelihood method and listed in Table IV.

As shown in Figs. 4 and 5, both the proposed method and the traditional correlation model in [3] can model the correlations among different weather variables, and the total correlation errors calculated by these two methods are close to one another. However, the accuracy of the correlations calculated by the two methods varies among the HVAC simulations. The correlation of the HVAC energy consumption at different buses will have a significant impact on the power grid operation. We simulated three total HVAC loads in three different temperature regions using different methods, and the total correlations were calculated using (15).

Note that the abscissas in Figs. 5 and 6 refer to the sample size of the Monte Carlo method for calculating the total correlation rather than the sample size of the uncertainties. From the point of view of convergence, (15) can accurately calculate the total correlation. It can be seen from Fig. 5 that the result of the proposed method is farther away than that of the model

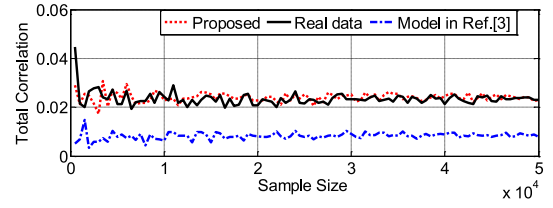


Fig. 6. Total correlation values of HVAC consumption in different zones.

in [3] from the real data. To explain this phenomenon, we need to make the similarities and differences between the proposed method and the model in [3] clear. The same point is that both of them use copula functions to model the cross-correlation, which is the essential reason for the above phenomenon. The difference is that the former considers the dynamic change law of correlation, but the latter does not. However, whether or not the dynamic correlation is considered will affect modeling the weather state transition probability and its correlation in Fig. 5. However, the dynamic change of the weather scenarios should be modeled using dynamic correlation, which is the basis of an accurate simulation of HVAC loads in Fig. 6. In other words, a dynamic correlation can improve the results in Fig. 6 rather than Fig. 5. It is noted that the accuracy of the proposed method is consistent with that of [3] in Fig. 5. It can be concluded that choosing the slightly better method depends on the stochastic Monte Carlo simulation results, and either method can have better results than the other. The difference between the proposed method the model in [3] brought by the theoretical correctness in Fig. 6 is greater than that difference brought by randomization in Fig. 5. By comparing Figs. 5 and 6, it can be concluded that the proposed method, which combines the theories of copula functions and Markov chains, agrees with the actual engineering requirements. The verification of the correlation among different loads is the theoretical basis for the following PPF simulation.

Discussion 2: PPF issues focus on the correctness of correlations among power generation and loads. From this point of view, an accurate correlation model is sufficient for PPF. However, with respect to weather-sensitive loads, the change law of weather scenarios should also be taken into consideration, and the Markov chain can accurately simulate the transition probabilities of different weather scenarios by controlling the dynamic correlations among different weather factors.

C. PPF Simulation

The simulation parameters of the HVACs in this section are the same as those in the above two sections. The load of each PQ node is set to a 1 kW base load plus 100 HVAC loads. To fully consider the impact of the correlation between the load and PV generation on the PPF, it is assumed that three 2 MW PV generations are installed at buses 3, 14 and 32 in case33bw. Using the weather simulation parameters in the previous section, we can obtain the simulation samples of the PV active power outputs and the HVAC loads. Monte Carlo is regarded as a benchmark for calculating the speed and accuracy of the proposed method. It should be noted that

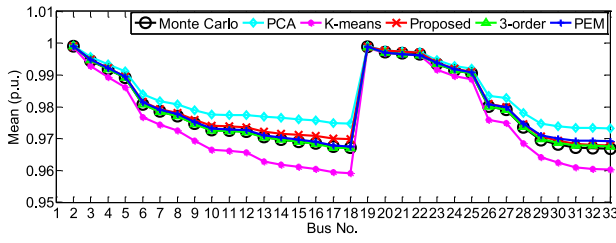


Fig. 7. Means of the PQ bus voltage (P.U.) in case33bw.

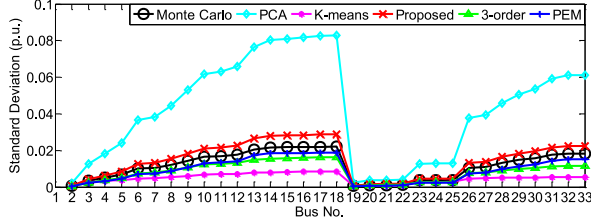


Fig. 8. Standard deviations of the PQ bus voltage (P.U.) in case33bw.

the data of the PV power and PQ node load used in this section are exactly the same for Monte Carlo and the proposed method. We simulate the PV power outputs and PQ node loads using the proposed method, where the sample size is set as 10000 to ensure the PPF calculation accuracy, i.e., we analyze and calculate the power flow distribution of 10,000 operation scenarios for case33bw. We then calculate the DPF of each typical scenario using MATPOWER.

To fully verify the validity of the proposed method, in addition to Monte Carlo, the principal component analysis (PCA), as in [8], and the k -means, as in [28], are also used to estimate the PPF, and their results are compared with those of this paper. With respect to the PCA method in [8], the threshold for a singular value is an important parameter that balances the computational efficiency and the solution accuracy of the PPF. The reduced sample size will be determined by the threshold. However, [8] does not provide a theoretical basis for calculating this threshold but rather gives an empirical value of 0.01, which is used in this paper. With respect to k -means in [28], it has been verified for the scenario reduction in [29]. To ensure fairness in the comparison, the number of typical scenarios in k -means is set to the sample size of the proposed method, which is equal to 56. From the simulation results, we can see that the reduced scenarios of the k -means method are generated using 66 iterations. Note that the power flow results of the k -means method are discrete variables with the same probabilities as those of typical scenarios, but the calculated power flows of other methods are continuous variables.

Computational efficiency and the solution's accuracy are two important indicators for evaluating the performance of the PPF. The means and standard deviations of the PQ bus voltages are estimated using different methods in case33bw, with the bus voltage profiles shown in Figs. 7 and 8. Similarly, the means and standard deviations of the total loss are estimated in case33bw, as shown in Table V. The CPU times of the PPF simulations are listed in Table VI.

To further elaborate on the advantages of the proposed method, the mechanisms of the different methods are compared and analyzed. First, the scenario-based method and

TABLE V
MEANS AND STANDARD DEVIATIONS OF THE TOTAL LOSS (MW)

Method	Mean	Standard deviation
Monte Carlo	0.08	0.03
PCA	0.09	0.05
k -means	0.07	0.01
Proposed	0.08	0.04
third-order	0.08	0.02
PEM	0.07	0.05

TABLE VI
CPU TIME AND DIMENSIONS OF THE PPF SIMULATION

Method	Scenario reduction (s)	PEM (s)	SRSM (s)	DPF (s)	Dimension
Monte Carlo	0	0	0	198.02	32
PCA	0.09	0	0.11	1.29	4
k -means	8.00	0	0	1.79	32
Proposed	8.80	0	0.14	1.79	6
third-order	16.86	0	1.53	1.79	6
PEM	0	0.15	0	1.89	32

Monte Carlo method are analyzed. Second, the comparison between the proposed method and the analytic method is analyzed.

The accuracy of Monte Carlo is the highest, but the calculation time is the longest. All scenarios are calculated using Monte Carlo, and the statistical results of the Monte Carlo simulation are used as benchmark and accurate values. The k -means method does not reduce the data dimension, but it reduces the number of scenarios. Although the CPU time was improved, this improvement caused a severe decrease in the accuracy. Both the PCA method and the proposed method reduce the scenarios along with dimensionality reduction, and they have a better accuracy than the k -means method. The difference between the PCA method and the proposed method is that the proposed method reduces the dimension of the input variables according to its intrinsic dimension, while the PCA method reduces the dimension of the input variables according to an empirical threshold. Obviously, the theoretical basis of the proposed method is better than that of the PCA method. In addition, the proposed method retains the nonnegative characteristics of the input variables while reducing the dimension; alternatively, the performance of the PCA method with respect to negative elements is poor.

Reference [30] demonstrated that a second or third-order polynomial is sufficient. When the SRSM polynomial order is greater than three, the increased accuracy can be negligible with an increasing order. Hence, a detailed comparison has been made between the second-order SRSM and the third-order SRSM in terms of the calculation time and accuracy. There are two differences between the legend "third-order" and legend "proposed" in Figs. 7 and 8 and Tables V and VI. The first difference is that polynomial order is two for the "proposed", and the polynomial order is three for the "third-order". The other difference is that the "proposed" uses (35) and (36) to calculate the mathematical expectation and variance, but the "third-order" needs to calculate the response vector using (31) to determine the mathematical expectation and variance. From the calculation results in

TABLE VII
STOCHASTIC PROGRAMMING RESULTS

Solution	Method	Scenario size	Time (s)	f_{obj} (MW)	α (%)
Lower boundary	First	50000	999.5	0.080	99.6
	Second	168	3.7	0.080	92.2
	Third	168	3.6	0.080	100
	First	50000	10050	0.222	95.8
Upper boundary	Second	168	3.7	0.223	93.7
	Third	168	3.6	0.223	100
	Second (First)	168×25×15	1427.5	0.031 (0.030)	0.998 (0.998)
	Third (First)	168×25×15	1404.7	0.027 (0.026)	0.998 (0.996)

Figs. 7 and 8 and Tables V and VI, it can be seen that the proposed method has a better performance considering both the calculation time and efficiency, and the second-order SRSM is feasible and practical in the PPF calculation. In summary, the third-order SRSM is more accurate than the second-order SRSM, and the cost is an increase in the time consumption.

PEM is regarded as an important and popular analytical method [31]. Thus, its results are compared with those of the proposed method to judge the advantage of the proposed method. The comparison between the PEM and the proposed method can be concluded as follows:

a) The PEM cannot deal with the dynamic correlation of uncertainties in distribution networks, but the proposed method can model the dynamic correlation via cross-correlation and state transition probability. The proposed method can study the PPF from the perspective of scenario theory, as it can simulate random HVAC loads and intermittent PV power considering weather scenario changes. However, it has difficulty dealing with the PPF dynamic correlation problem using the PEM. The dynamic correlation of the input variables has significant impacts on the statistics of the power flow responses, so from this point of view the proposed method is better than the PEM.

b) From the calculation results in Figs. 7 and 8 and Tables V and VI, it can be seen that the proposed method has no better performance over the PEM considering both the calculation time and efficiency. Of course, the above phenomenon is reasonable. The analytic method has natural advantages over the scenario-based method in terms of calculation time and accuracy. However, it should be noted that the usability of the PPF analytic methods depends on the correctness and refinement of mathematical model, especially the DPF model. This is not a problem in simulations, but it is a problem in terms of practical engineering. In China's power grid, many impedance parameters of the distribution networks have not been correctly verified. Errors in the DPF parameter lead to an uncertainty analysis error regarding the physical model [7]. In real applications, we can select the grid measurement data under different scenarios instead of the DPF simulation data. It can be considered that the scenario-based method is data-driven. However, real data collection may be unavailable for analytic methods. It is thus impossible to collect the

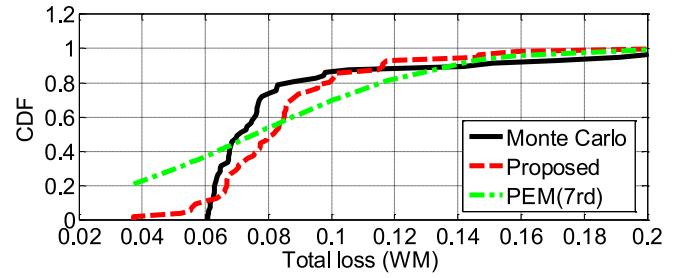


Fig. 9. CDFs of the total losses for different methods.

state variables under given operation conditions, because the experimental operation conditions obviously do not exist in the actual power grid. For example, the input variables expect one variable to be set to the mean when we use the PEM to estimate the power flow responses, and this condition is an assumption rather than real. In conclusion, no DPF parameter error is the premise of an excellent analytical method. The proposed method is based on statistical machine learning of the scenario data, which can avoid a dependence on complex physical parameters, and is suitable for scenario modeling of power grid planning.

c) The probability distributions of power flow responses cannot be obtained accurately and directly using the PEM. To obtain the CDF, first we used the PEM to obtain the origin moments; second, we compute the cumulants according to the relationships between the cumulants and the origin moments using (12) in [32]; Finally, we compute the CDF of the power flow responses using Gram-Charlier expansion formulas in [33], and the Gram-Charlier coefficients are calculated using cumulants. Generally, taking the first seven order semi invariants can ensure a high accuracy. However, the convergence of the Gram-Charlier expansion cannot always be ensured. In contrast, the proposed method can produce available power flow responses, whose CDF can be accurately and directly estimated via Kaplan-Meier estimation. As shown in Fig. 9, the curve of the PEM with the seventh order Gram-Charlier expansion is significantly biased from that of the Monte Carlo method, and the curve of proposed method is close to that of the Monte Carlo method.

Discussion 3: Large datasets and data consisting of a large number of features present computational problems in the PPF calculation in distribution networks. From the perspective of machine learning, k -means belongs to the clustering method, and the proposed method belongs to the feature dimensionality reduction method. The former reduces the number of data points, and the latter reduces the amount of data scale and dimensions. The scenarios of state variables corresponding to different scenarios of input variables may be consistent. In the k -means algorithm, the probabilities of state variables and random input variables are considered to be identical; this assumption leads to more PPF errors than the other two methods.

D. Stochastic Programming Simulation

Calculating the DPF in an ocean of scenarios is time consuming, which makes it impossible to solve a feasible solution of stochastic programming in a given time; this is why we need

a fast PFF calculation method. The importance of modeling for programming is of vital importance. A good programming model may be more effective than a good solver. To clarify the model and solver clearly, we first analyze the availability of the programming model, and then analyze the solver performance.

With respect to the programming model, here we focus on the confidence level of the voltage inequality constraints. The system parameters before planning are exactly the same as those in the above section; the difference is that a more accurate third-order SRSIM is used here. We plan capacitors at buses 14 and 32, and the range of the decision variables is [0 kvar, 2000 kvar]. The qualified voltage range is [0.9 p.u., 1.1 p.u.] at buses 14 and 32, and the confidence level of the constraint is set to 99.8% for the programming model. We use three methods to calculate the constraint confidence levels of the stochastic programming model. The first is the interval statistics of the Monte Carlo results; the second is Chebyshev inequality estimation of the SRSIM responses; and the third is the interval statistics of SRSIM responses. It should be noted that the Monte Carlo method is unavailable for the programming model because it is too time-consuming. The optimal solutions of the second and third methods are checked by the Monte Carlo method to obtain the accurate objective function and constraints. We use Monte Carlo as a benchmark to verify the validity and rationality of the Chebyshev inequality. In addition, it should be noted that 25×15 denotes the particle population size multiplied by the number of iterative steps, which are the solver parameters. The simulation results of the stochastic programming are listed in Table VII.

The response interval statistics method may be optimistic, while the Chebyshev inequality estimation method tends to be pessimistic. Although the objective function of the third method is more excellent than the second method, its solutions cannot meet the constraint conditions. With respect to reactive power planning optimization, it is reasonable to be conservative. More importantly, the conservative nature of the Chebyshev inequality before planning is greater, but the conservative nature of the Chebyshev inequality after planning is reduced. The optimal solutions and constraints based on the Chebyshev inequality are nearly the real values. The reason for the above phenomenon is that when ε decreases after optimal planning, the confidence level of the Chebyshev inequality becomes more accurate.

Now let us discuss the solver, which is a part of the reactive power planning work. It must be stated that this paper does not focus on the performance of the solver but instead on the impacts of the fast PFF calculation methods on the solving efficiency of the programming model. Note that the mathematical programming solver is not a contribution of this paper, and the solver used in this paper can be replaced with a different solver with a better convergence. Convergence is one of the most important evaluation indexes of the evolutionary algorithm, and we can choose a solver whose convergence has been judged. Reference [34] proved that the quantum-behaved particle swarm optimization (QPSO) is a global convergence algorithm, and the SPSO can reach the global optimum in terms of probability. The QPSO is used to solve the problem

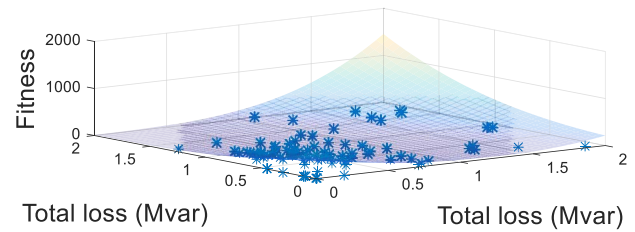


Fig. 10. Particle positions in the QPSO optimization.

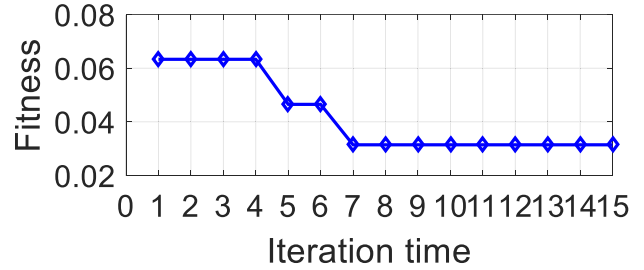


Fig. 11. Fitness value of global optimal solution via iteration times.

of the reactive power planning optimization. The parameters of the solver directly affect the accuracy of the result, and we choose the validated QPSO parameter calculation formulas in [35] to ensure the effectiveness of the optimization solution. The solution process of the statistical machine learning model for stochastic optimal planning of distribution networks is drawn to show the QPSO convergence performance. As shown in Figs. 10 and 11, it can be concluded that QPSO has a strong randomness and covers most solution spaces on the bottom to ensure global convergence of the algorithm.

Discussion 4: The level of accuracy achieved when solving a power planning problem with a stochastic programming model largely depends on the approximation degree between the stochastic model and the actual complex scenario. How to properly compute the probabilistic distributions of the state variables so that they approach the real values in a reasonable amount of time is the focus of this research. The proposed method solves the problem of constraint satisfaction via Chebyshev inequality theory based on the PPF results, ensuring the efficiency and accuracy of stochastic optimal planning in distribution networks.

VII. CONCLUSION

Statistical machine learning is a science concerned with data-based probabilistic statistical models that are constructed by computers and used to predict and analyze data. It is not only the core problem in the field of artificial intelligence but it has also become one of the most active research branches in the field of electrical engineering. Intermittent renewable generators and stochastic electricity use behaviors that clash with distribution networks with complex randomness, making the stochastic programming model very difficult to solve. In light of the above stochastic programming problems, a novel stochastic programming method is proposed based on statistical machine learning theories, including digital characteristics, correlations, nonnegative matrix decompositions, stochastic

response surfaces, and Markov chains. The proposed method can quickly calculate probabilistic power flow and establish a functional relationship between the probabilistic power flow and stochastic programming. The simulation results show that the proposed method can help find the optimal planning scheme of distribution networks that must consider a large number of stochastic scenarios in a given amount of time.

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